CAPSTONE STUDY GUIDELINES
OF THE COLLEGE OF INDUSTRIAL TECHNOLOGY

I. OBJECTIVES:

A capstone study is a major academic requirement for graduating students in both undergraduate and graduate programs. It aims to measure the comprehensive understanding, analysis, evaluation, and practical application of knowledge, experiences, and skills of students in solving real-world problems relative to their major field of specialization. It involves key processes such as needs assessment, deep research, design, development/ fabrication/ formulation, testing/ analysis, and/or evaluation stages, with the main goal of producing tangible and high-impact output, which can be in the form of a project, prototype, system, or product per group (undergraduate) or per student (graduate). These outputs are expected to potentially add value to adopters, communities, industries, including the scientific and academic communities, and other stakeholders.

This capstone study guidelines aim to serve as a blueprint in preparing and writing a concise documentation of the manuscript of the students. It features a simplified approach of capturing the salient information based on best practices of knowledge and innovation generation, protection, diffusion, and commercialization processes, as applicable. Moreover, this set of guidelines is subject to continual improvements based on the suggestions and recommendations of experts, concerned faculty members, and higher authorities.

II. SCOPE:

This set of guidelines shall be applicable to all students in both graduate and undergraduate programs of the College of Industrial Technology. It may be used as a reference by other colleges, campuses, and extension programs of the Technological University of the Philippines (TUP), through a written approval by the Dean of the College of Industrial Technology.

III. RESPONSIBILITIES:

The students should adhere to this capstone study guideline in preparing and writing their manuscripts.

The advisers should provide direct advisory and mentorship in ensuring that the manuscript complies with the guidelines and toward the successful realization of the student/s' actual project/product/prototype/system.

The faculty Members sitting as Panel Members should provide critiquing, technical assistance, insights, evaluation, and recommendations to their assigned capstone studies.

The University Research and Development Services (URDS) is enjoined to provide technical assistance toward the issuance of a certificate of similarity rating or equivalent, based on the submitted manuscripts of the students.

The Proof Readers/ Editors are enjoined to provide quality and cost-effective proofreading and editing services for the manuscripts of the students.

The Professors of Research and/or Project Development Courses should directly mentor and guide the students in the preparation and writing of the manuscripts, without prejudice to their academic freedom.

The Department Heads should monitor the compliance of the students in the guidelines, in coordination with the assigned Professor, advisers, faculty members/panel members, the Dean of the college, and other relevant interested parties.

The College Dean should ensure the effective implementation of this set of guidelines, in coordination with all relevant interested parties at the college and/or university level, including external parties and regulatory bodies, as applicable.

IV. SPECIFIC GUIDELINES

This set of guidelines aims to document concisely the manuscript of a capstone study based on the chronological order of the following key sections or sub-sections with examples:

1. PRELIMINARIES

1.1 Title Page - This page contains the following sections:

- **Title –** The title of the capstone study should be placed four lines from the top of the page. The title should be in upper case, boldface, single-spaced, and in inverted pyramid format.

- **Affiliation –** This details the name of the department, college, university, and address. The name of the department should be in boldface.

- **Student/s –** This contains the name/s of the student/s where the surname/s is/are alphabetized; however, the format of the name/s should be first name, middle initial, and last name. The name/s should be in upper case and boldface. The order may be based on the weight or degree of contribution/s, or as agreed among the group members.

- **Program** – This contains the name of the student/s' program and major field of specialization.

- **Date** – This contains the month and year of the final defense.

The spacing between each line of the Title Page should be proportional to ensure balance and cohesiveness.

Example:

TITLE OF THE CAPSTONE STUDY
(FONT SIZE: 14)

A Capstone Study (Font Size: 12)

Presented to the Faculty of the
NAME OF THE DEPARTMENT
College of Industrial Technology
Technological University of the Philippines
Ayala Boulevard, Ermita, Manila

by

FIRST NAME MI. LAST NAME
FIRST NAME MI. LAST NAME
FIRST NAME MI. LAST NAME
FIRST NAME MI. LAST NAME
(FONT SIZE: 12)

In Partial Fulfillment of the Requirements for the Degree
Name of Program

Month, Year of Final Defense

1.2 Approval Sheet - This contains the heading of the college, the title of the capstone study, the student/s' name/s, and the program. The approval sheet should be duly signed by the advisory committee and the Dean before bookbinding the final manuscript.

Example:

Technological University of the Philippines
COLLEGE OF INDUSTRIAL TECHNOLOGY
Ayala Boulevard, Ermita, Manila 1000
cit@tup.edu.ph

APPROVAL SHEET

The capstone study titled "**TITLE OF THE CAPSTONE STUDY**", prepared and submitted by **NAME/S OF STUDENT/S**, in partial fulfillment of the requirements for the degree **Name of the Program**, is hereby approved and accepted.

FIRST NAME MI. LASTNAME
Adviser

FIRST NAME MI. LASTNAME
Chairperson

FIRST NAME MI. LASTNAME
Member

FIRST NAME MI. LASTNAME
Member

FIRST NAME MI. LASTNAME
Member

FIRST NAME MI. LASTNAME
Member

FIRST NAME MI. LASTNAME
Dean, College of Industrial Technology
Date: _____

1.3. Table of Contents - This lists the contents of the manuscript, including the page for each section and sub-sections.

Example:

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1.4. List of Tables - This lists the tables as discussed in the manuscript.

Example:

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1.5. List of Figures - This lists the figures as discussed in the manuscript.

Example:

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1.6. List of Abbreviations - This lists the definition of all pertinent abbreviations or acronyms used in the capstone study. It should be arranged alphabetically/ chronologically.

Example:

LIST OF ABBREVIATIONS

Abbreviation	**Definition**
Abb1	Abbreviation meaning here
Abb2	Abbreviation meaning here

1.7. Abstract with Keywords – This is a concise narrative of the contents of the paper, consisting of 200 to 250 words, written in one paragraph only. It includes the discussion of the following: main topic, problem, research gap, objectives, methods, results, conclusions, and implications of the study, e.g., alignment with various research agenda/UN-SDGs, or impact on the beneficiaries. The abstract is required for the final manuscript only. The keywords are terms or phrases used to categorize the study for indexing or searching purposes. Keywords can be in the context of the main topic, technology, area of study, etc.; the capstone study should have at least three to five keywords based on their degree of relevance.

Example:

ABSTRACT

Electric bicycles (E-bikes) are gaining traction in the transportation industry due to their accessibility, affordability, and eco-friendly nature. Overall, they are convenient to use and significantly reduce carbon footprints. However, they have limited battery capacity and limited support infrastructures, which hinder long-distance travel. Existing solutions, like conventional or solar-powered charging stations, have drawbacks such as being time-consuming, manually operated, and the need for monitoring and supervision. The researchers addressed these opportunities by designing and developing an Internet of Things (IoT)-enabled E-Bike charging-vending machine with a lock and location tracking features to improve user convenience and functional efficiency. The researchers utilized a prototyping approach to attain the research objectives by leveraging IoT to achieve the anticipated solution. Then, the researchers tested and analyzed the system in terms of timeliness and functional efficiency based on hypothesis testing techniques. Test results showed that the system is significantly faster than the conventional method in terms of its pay-charging feature ($t(34.740) = -8.786$, $p < 0.001$). On the contrary, the researchers found that there is no significant difference in the functional efficiency based on the measured voltage rating using the digital tester as compared to the data displayed on the mobile application ($t(28) = -0.035$, $p = 0.486$). The results proved that the displayed voltage rating on the mobile application is at par with the measured voltage rating using the digital tester. This innovation benefits the electric bicycle owners and the community in general by contributing to an enhanced operational efficiency and, more importantly, in reducing carbon footprint on a daily basis. The outcomes of this study are aligned with the United Nations' Sustainable Development Goals (UN-SDG) 7, 12, and 13.

Keywords: E-bike Charging Station, IoT, RFID-Based Vending Machine, UN SDG 7, UN SDG 12, UN SDG 13

2. MAIN SECTIONS and SUB-SECTIONS

I. INTRODUCTION

1.1 Background of the Study

This section describes the background of the capstone study, based on the main research topic and problem that needs to be addressed. The following is the suggested flow:

- The first paragraph should concisely discuss the main topic, which is the specific area, technology, phenomenon, or field of study of the capstone study.
- The second paragraph should concisely discuss the main problem that the capstone study intends to solve.
- The third paragraph should concisely describe the causes of the problem.
- The fourth paragraph should concisely highlight the research gap, which is the opportunity for improvement on existing prototypes/projects/products/systems that may serve as references in proposing potential solutions to the problem.
- The fifth paragraph should concisely discuss the best and most feasible solution or strategy to solve the problem.

The introduction should be supported with extensive conceptual and empirical literature. At least three (3) authors should be cited per paragraph, which should be taken from refereed sources only, i.e., research articles from published journals, books, book chapters, conference proceedings (preferably indexed in Scopus, Web of Science, ASEAN citation index), government websites, patent databases, and reputable non-government organizations (e.g., UN, UNESCO, WHO, etc.).

Example:

Hospitals are considered the backbone of healthcare systems around the world. They play a crucial role in maintaining good public health by providing immediate care to sick or injured people; isolation facilities or venues for medical research; and other support functions to the government and communities (Nilsen et al., 2020; Mass et al., 2009). One of the measures of quality of hospital services is the satisfaction of patients, primarily with the availability of hospital beds. In one study by Xesfingi and Vozikis (2016), they found that there is a negative association between the number of hospital beds vis-à-vis the satisfaction of patients. Moreover, in another study by Asamrew and Tadesse (2020), they found that the availability of hospital beds is a crucial predictor of patient satisfaction. Interestingly, the ultimate root cause of bed shortage in hospitals is due to overcrowding, which is a common dilemma in most hospitals (Soyomei and Aborde, 2022; Behadoria et al., 2022), particularly during pandemics (Savioli et al., 2022).

Overcrowding in hospitals happens if there is a huge number of incoming patients; there is a delay in admission, and for worse, premature deaths (Tampubolon and Pujiyanto, 2018). Overcrowding can be further attributed to various factors, such as poor bed management (Jarvis, 2016); issues in supply chain management (Stansbury et al., 2021); poor coordination (Rojas-Garcia et al., 2021); limited manpower (Di Somma et al., 2015); lack of a sophisticated system to monitor real-time data (Salway et al., 2017; Schmidt et al., 2017); inefficient decision-making process (Noonan et al., 2019); changing decisions of patients to stay (Hoffman et al., 2019); and prevalence of communicable diseases in hospitals (Silva et al., 2020), among others. In the Philippines, there are fewer public hospitals than private ones in the entire archipelago, which results in more than 50% of the population who cannot afford basic healthcare services (Flores et al., 2021). Recently, during the pandemic, many Filipino families struggled to find available hospitals even in nearby towns, cities, or provinces due to overcrowding. This resulted in higher death rates and brought significant distress to patients, hospital managers, health authorities, and local and national government officials (Abedian et al., 2018).

Fortunately, several technological innovations were developed to address the problem of hospital bed management. For instance, Abedian et al. (2018) developed a real-time, web app with a trigger feature; however, it lacks a warning system for the shortage of ICU and CCU beds, which may result in an influx of ICU and CCU patients. Abedian et al. (2014) also developed another web-based bed management system with a referral feature; however, it does not have a provision to add new beds. In another study by Ajmi et al. (2022), they utilized an IoT beacon with mobile and web apps. However, it has some concerns in location tracking and data security. In another study by De Morais et al. (2020), they developed a web-based ICU bed information system with a comprehensive dashboard; however, it lacks an electronic medical records-archiving feature and patient regulation system. Lastly, Cahyo and Utaminingrum (2022) co-developed an autonomous robot that detects hospital room nameplates using the YOLOV4 method, which is deemed 94% accurate. However, its movements are limited in tight spaces, and it has a buzzing sound that creates unnecessary noise.

Moreover, numerous technological inventions have been patented that aim to monitor hospital beds. In the work of Salgo and Golden (2008), they utilized a pressure-sensing system,

display monitor, and alarm system to monitor available hospital beds. While the said system is informative and functional overall, it is found to be efficient; it requires numerous computer peripherals such as a handheld Personal Digital Assistant (PDA) and a nursing station computer, hence, it is expensive to set up. Meanwhile, DiMaio et al. (2013) also patented an integrated patient bed system using various diagnostic tools and equipment such as fluoroscopy, x-rays, CT-scan, ultrasound, cardiac monitor, dialysis machine, blood analysis, etc., to monitor critically ill patients. While the system is comprehensive in data generation, it is costly and complicated to manage.

Overall, these technological developments and patents are deemed efficient, real-time, and sophisticated. However, further work is needed to explore alternative and sustainable solutions to optimize the monitoring of hospital beds through simplified and decentralized management of data through continuous innovation of feature sets and the Internet of Things (IoT) framework (Kravets et al., 2024). In this current study, the researchers proposed a monitoring system that can provide vital information on the availability of hospital beds by leveraging on IoT, hence, potentially lessening the cases of overcrowding and effectively managing hospital beds.

1.2 Objectives of the Study

General Objective

The general objective captures the main intent of the capstone study. It describes the qualitative outcome expected in the context of the beneficiaries. It should answer the question "How will the intended stakeholders benefit from the prototype/project/product/system?"

Specific Objectives

The specific objectives must be explicit and precise, and the expected results must be quantitative and verifiable. The criteria of being specific, measurable, attainable, realistic, and time-bound (SMART) must be considered in setting the specific objectives. The following are the suggested process-based, specific objectives:

For Prototype/Project/System Development

a. Design – specify the core technology to be used, including the hardware and software features, functionalities, or solutions
b. Fabricate/Develop– specify the framework or methodology to be used in translating the design into a tangible output, such as prototyping, waterfall, agile, scrum, V model, etc.

c. Test – specify what technical characteristics are to be measured, analyzed, or tested; e.g., accuracy, timeliness, efficiency, reliability, etc. Ideally, testing should be proven using various statistical techniques or international standards.

For Food/Product Development

a. Formulate – specify the ingredients of the food/ product, including the mix ratio, etc.
d. Develop – specify the framework or methodology to be used in translating the formulation into a tangible output, such as stage gate food development, experimental, agile, HACCP, mixed methods, etc.
e. Analyze – specify the food characteristics to be analyzed; e.g., nutritional, physico-chemical, microbiological contents, etc. Ideally, the analysis should be proven using various statistical techniques or international standards

For Material Science/ Construction Technology Research

a. Characterize – specify the technical, structural, or morphological properties of the material/s
b. Formulate/ Develop – specify the mix design or the methodology to use, e.g., experimental, prototyping, engineering process, etc.
f. Test – specify the technical characteristics to be measured, e.g., mechanical, thermal, or electrical properties, durability, corrosion resistance, including the effects on varying the material mix, etc. Ideally, the testing should be proven using various statistical techniques, or international standards, e.g., ASTM, PNS, etc.

Example:

This capstone study aims to support the healthcare system in the country through the provision of a real-time, efficient, and accurate hospital bed management system. Specifically, the researchers proposed to achieve the following objectives:

1. Design a hospital bed monitoring system with the following features:
    a. Web application for patients and the public
    b. Desktop application for hospital administrators
    c. An IoT-powered detection system for the availability of hospital beds using reflective optical sensors, a microcontroller unit, a Wi-Fi module, and system peripherals.
2. Develop the system using the agile scrum methodology.
3. Test the system in terms of functional accuracy through a hypothesis-testing approach.

1.3 Scope and Limitations of the Study

The scope discusses the extent and boundaries, particularly in the high-level context as to why and how the study was conducted or made. The major methodologies of designing,

developing, and testing/analyzing the prototype/project/product/system processes should be described. Meanwhile, the limitations are the factors that are not covered by the study, such as the applicability of the capstone study, procedures not covered, sampling constraints, its functionalities or capability features, technical characteristics, intended users, etc.

This section also identifies the specific beneficiaries and how they will benefit from the outcomes, including the alignment of the capstone study with national and international research or innovation agenda (e.g., TUP, DOST, and UN-SDG).

Example:

This capstone study primarily aims to design, develop, and test a web and desktop applications, and a hardware prototype to monitor the availability of hospital beds in the country. The researchers utilized the Internet of Things (IoT) technology and locally available materials, such as Arduino Uno Microcontroller Unit (MCU), ESP8266 ESP-01 WiFi module, CNY70 reflective optical sensors, 4-piece 1.5V batteries, and other system peripherals. They developed both the software and hardware requirements using Agile Scrum methodology.

For the testing, the researchers analyzed the performance of the system in terms of accuracy and timeliness, in comparison with the conventional method, through the Analysis of Variance (ANOVA) and hypothesis testing techniques. Moreover, they involved health care and IT administrators and staff as pilot respondents and obtained their feedback and recommendations toward the improvement of the system.

However, the system has some technical limitations. For example, the desktop application requires a stable internet connection, and the administrator's computer hardware should have at least 4 cores and be in the same local area network. Meanwhile, the hardware prototype can only detect white or light nameplates. It should also be installed in the same area as the server, and it has a limited battery source.

This capstone study is expected to be beneficial to the general public, who are provided with general information and an alternative method of finding available hospitals for their medical needs. It is also valuable to the hospitals' staff and administrators, who are provided

with a support system to efficiently manage their bed availability and lessen patient flow in hospital facilities. Moreover, the Department of Health (DOH) and other health care authorities can be provided with general information on the demographics, trends, and preferences of the general public, including the performance of hospitals in terms of hospital bed availability and management. These insights can be used for future policy and decision-making purposes. Lastly, the academia and researchers are provided with technical results and scientific data, which can be used as a reference for future development, social, or theoretical research.

This capstone study is aligned with the TUP research agenda on "Universal solutions for food security and health" and on the United Nations' Sustainable Development Goals (SDG) on Good Health and Well-Being (Goal 3).

II. LITERATURE REVIEW

This section discusses the details of the **main topic**, the **problem** and its root causes, and other pertinent information that provides deep theoretical or conceptual and practical underpinnings of the capstone study. Students should cite at least five (5) refereed sources each for the main topic and problem based on similar context. The use of various frameworks, such as cause-effect analysis, fishbone, Ishikawa chart, etc., is recommended.

Moreover, this section highlights the **research gap** or the justification as to why the capstone study should be conducted. It discusses a comprehensive review of pertinent theoretical, conceptual, and empirical literature and the analysis of the strengths and weaknesses of existing studies, e.g., published research papers and patents that are **directly similar** to the proposed prototype/project/product/system.

To elaborate on the research gap, the following questions should be answered: (1) What are the strengths and weaknesses of the existing technologies, patents, or solutions? (2) What makes the current study new, novel, and original? And (3) how is the current study worth contributing to the scientific body of knowledge? Students should cite at least ten to twenty (10 to 20) prototypes/projects/products/systems that are directly similar to the current study, and five to ten (5 to 10) directly related patents, utility models, and other relevant intellectual property rights. The research gap can be presented in various frameworks or strategies, such as Strengths-Weaknesses-Opportunities-Threats (SWOT) analysis, cost-benefit analysis, comparison matrix, etc. Take note that **only refereed sources are allowed**, i.e., research articles from published journals, books, book chapters, conference proceedings, preferably indexed in Scopus, Web of Science, ASEAN citation index, government websites, patent databases, reputable non-government websites (e.g., UN, UNESCO, WHO, etc.). **Random literature, figures, and tables taken from the internet are not allowed.**

The following are the sample formats for citing figures and tables, which are used extensively in the Literature Review and Methodology sections. These should be legible, clear, and easy to interpret. A note can be used to further explain the figure or table concisely, then cite the original author appropriately.

Table #

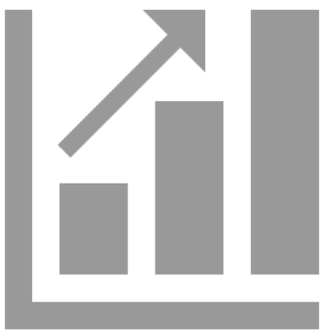
Title of Table

Column 1	Column 2	Column 3	Column 4
Data 1-A	Data 2-A	Data 3-A	Data 4-A
Data 1-B	Data 2-B	Data 3-B	Data 4-B
Data 1-C	Data 2-C	Data 3-C	Data 4-C
Data 1-D	Data 2-D	Data 3-D	Data 4-D
Data 1-E	Data 2-E	Data 3-E	Data 4-E

Note. A note can be used to describe the insights on the table; directly cite the author/s using the APA format (Author/s, year) if the table is used in the literature review section.

Figure #

Title of Figure



Note. A note can be used to describe the insights on the figure; directly cite the author/s using the APA format (Author/s, year) if the figure is used in the literature section.

Example:

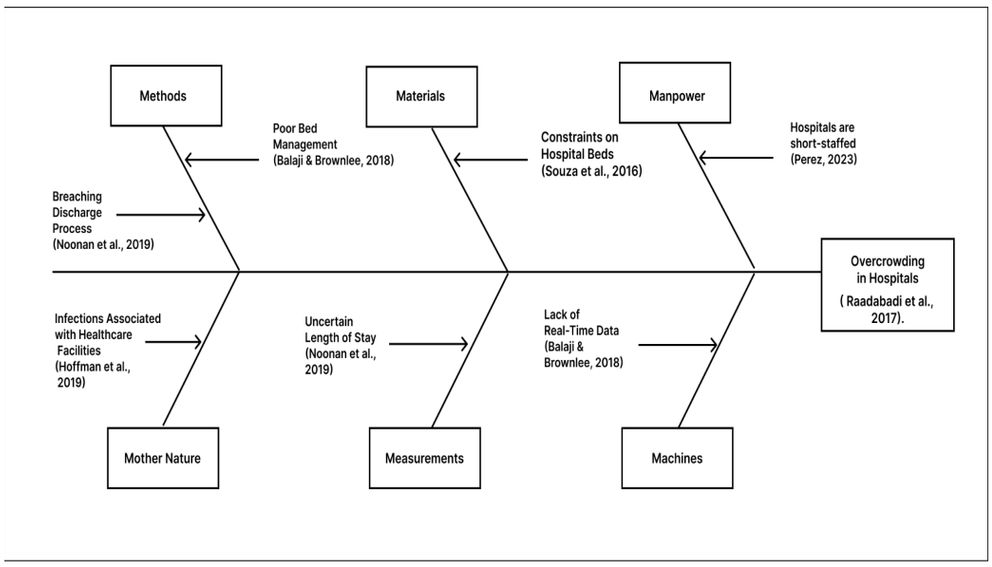
Hospitals play a crucial role in ensuring the health and wellness of people. It is a core foundation of health care systems as they are used for various purposes. They provide immediate care to sick or injured people. They serve as isolation facilities or venues for medical research, and other support functions to the government and communities (Nilsen et al., 2020). Over the years, operations in both public and private hospitals have evolved to meet the growing needs of patients and clients (Mass et al., 2009). As such, hospitals are expected to level up their performance through the provision of quality and equitable medical services and the integration of emerging technologies to reform their operations (Najibi et al., 2022).

Overcrowding is one of the main problems in hospitals and clinics (Raadabadi et al., 2017). Hospitals are working to increase their outpatient offerings to shorten the length of stay of their patients due to financial constraints and implications (Cayirli and Veral, 2009). As such, it negatively impacts the patients' and employees' access to services, schedule overlapping, lengthy waiting times, and patient and staff dissatisfaction (Parks et al., 2011). The following Figure 1 shows the cause-and-effect model, which illustrates the root causes of the research problem, i.e., overcrowding in hospitals (Raadabadi et al., 2017). As shown, the major root causes include the lack of manpower (Perez, 2023), hospital bed constraints (Souza et al., 2016), poor bed management (Balaji and Brownlee, 2018), inefficient discharge process (Noona et al., 2019), hospital infections or pandemics (Hoffman et al., 2019), uncertainty on length of stay (Noonan et al., 2019), and lack of real time data (Balaji and Brownlee, 2018).

One of the measures of quality of hospital services is the satisfaction of patients, primarily with the availability of hospital beds. In one study by Xesfingi and Vozikis (2016), they found that there is a negative association between the number of hospital beds vis-à-vis the satisfaction of patients. Moreover, in another study by Asamrew and Tadesse (2020), they found that the availability of hospital beds is a crucial predictor of patient satisfaction, and the ultimate root cause of overcrowding, a common dilemma in most hospitals (Soyomei and Aborde, 2022; Behadoria et al., 2022), particularly during pandemics (Savioli et al., 2022).

Figure 1

The Cause-And-Effect Model



The researchers conducted an extensive literature review to determine the strengths, weaknesses, opportunities, and threats (SWOT) of existing technological solutions, which served as a reference in the design and development of the current capstone study. Shown in Table 1 is the summary of the SWOT analysis, taken from published refereed sources.

Table 1

SWOT Matrix

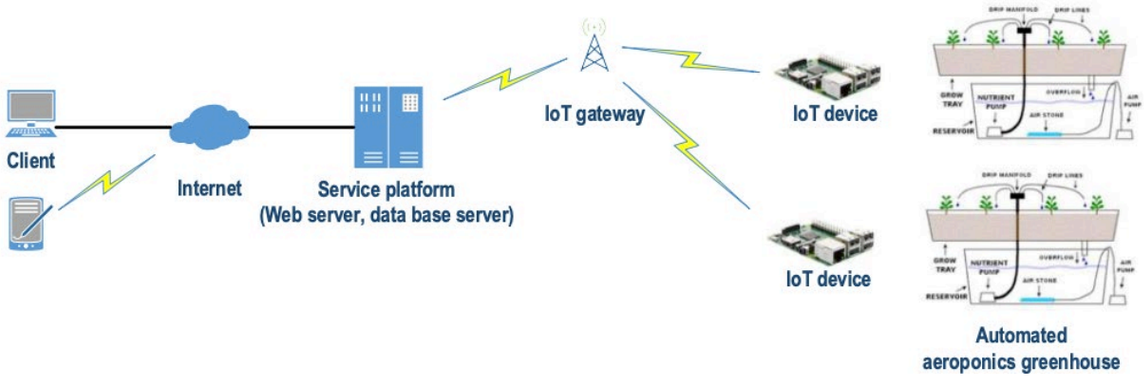
Research Title (Author)	*Methodology*	*Strengths*	*Weaknesses*	*Opportunities*	*Threats*
Automated Aeroponics System Using IoT for Smart Farming (*Kerns & Lee, 2017)*	Mobile application, web and database servers, IoT devices, and sensors	Efficient, highly accessible interconnected devices	Absence of external power supply, needs internet	Install solar panel or alternative energy source	Operational inefficiency or system failure

Automated Aeroponics System for Indoor Farming using Arduino *(Rahman et al., 2018)*	Microcontroller with sensors and an IoT module	Independent urban indoor farming	Lack integration of pH sensor and cooling system; needs internet	Implement a pH sensor and cooling system	Rise of pH and temperature affect the growth of crops
IoT – Based Automated Aeroponics System *(Francis et al., 2018)*	Microcontroller, multiple sensors, and the data is sent to a cloud database through IoT	Interconnected devices through the internet; accessible database	No CO2 sensor	Implementation of CO2 sensors and artificial intelligence (AI).	Unmonitored level of CO2 and the system affects the metabolism of the crop.
Monitoring and Control of Aeroponic Growing System for Potato Production *(Idris & Sani, 2012)*	Monitoring and control system using multiple sensors	The nutrient solution and water circulate in the chamber; multiple sensors data	Requires an operator to change the environment of the specific crop	Use of AI	The chamber is limited to potato, hence, unreliable for other crops
Automatic Monitoring and Control System in Aeroponic Plant Agriculture *(Rahmad et al., 2020)*	Monitoring and control system of 3 sensors	Monitors multiple parameters of the aeroponic system	Limited on water spinach; Inefficient irrigation	Install a specialized irrigation system	Operational and maintenance cost

Shown in Figure 3 is the automated aeroponic system developed by Kerns and Lee (2017). It features a mobile application, web application, and database, integrated with sensors and the IoT architecture. Overall, the system is found to be efficient and highly accessible due to the interconnected devices of the IoT framework. However, it lacks an external power supply and requires a steady internet connection. Areas of improvement include the installation of green energy sources such as solar power; otherwise, it would result in operational inefficiency or potential system failure.

Figure 3

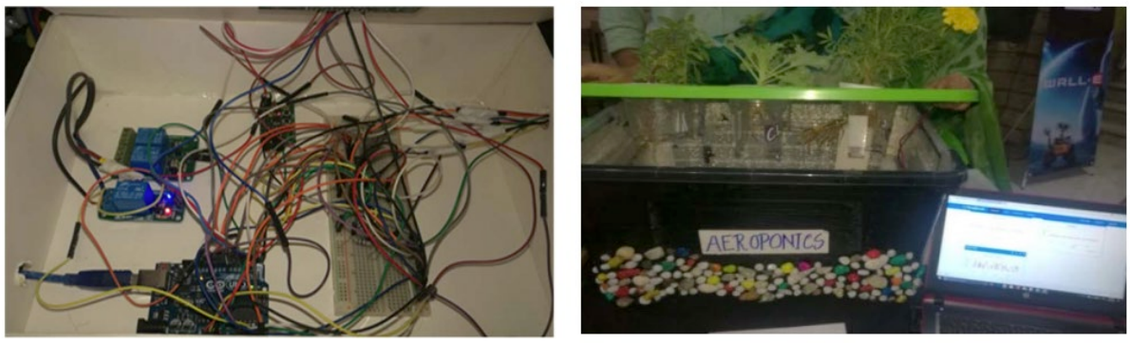
Automated Aeroponics System Using IoT for Smart Farming (Kerns & Lee, 2017)



In another related study, Rahman et al. (2018) developed an automated aeroponics system, as shown in Figure 4, using Arduino Uno, WiFi module, RTC module, LDR, and multiple sensors such as humidity, temperature, and sonar. Overall, their system is an efficient and independent aeroponic system designed for urban indoor farming. However, it lacks a pH sensor and a cooling system, which may affect the growth of the plants.

Figure 4

Automated Aeroponics System for Indoor Farming using Arduino (Rahman et al., 2018)



Then, Francis et al. (2018) developed an IoT-based automated aeroponics system, as illustrated in Figure 5. It uses multiple sensors, such as humidity and temperature, analog pH, integrated through a Raspberry Pi 3 microcontroller and Zigbee transceiver. While it features a manual and automated monitoring of sensor data for an efficient monitoring system, unfortunately, it lacks a CO2 sensor, which is crucial for a more accurate detection, essential for the growth of crops. Likewise, AI is recommended due to the potential of the multiple sensors and the capacity of the microcontroller used.

Figure 5

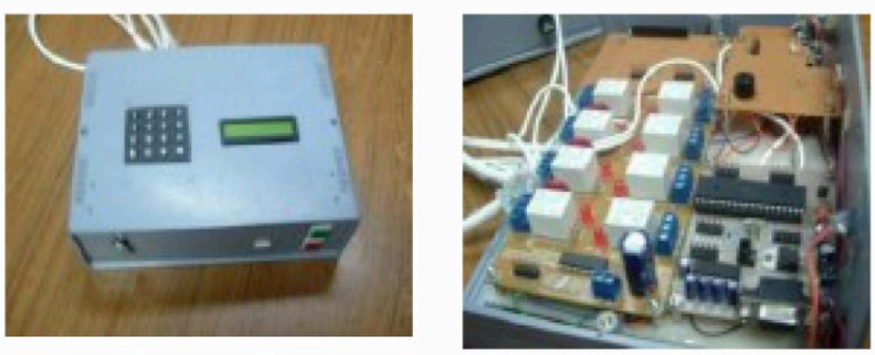
IoT – Based Automated Aeroponics System (Francis et al., 2018)



Idris and Sani (2012) also developed a monitoring and control aeroponic system for potato crops, as shown in Figure 6. The system features multiple sensors, such as pH, temperature, and humidity, connected to a microcontroller system, LCD display, and nozzle actuator in a growth chamber. However, the system requires a manual method and an operator to constantly monitor the status of plant growth, temperature, light, and humidity. Likewise, the system is limited to potato crops; hence, its applicability with other plants was not verified or established. Due to the use of multiple sensors, AI integration is recommended to enhance its flexibility in managing multiple data and crop types.

Figure 6

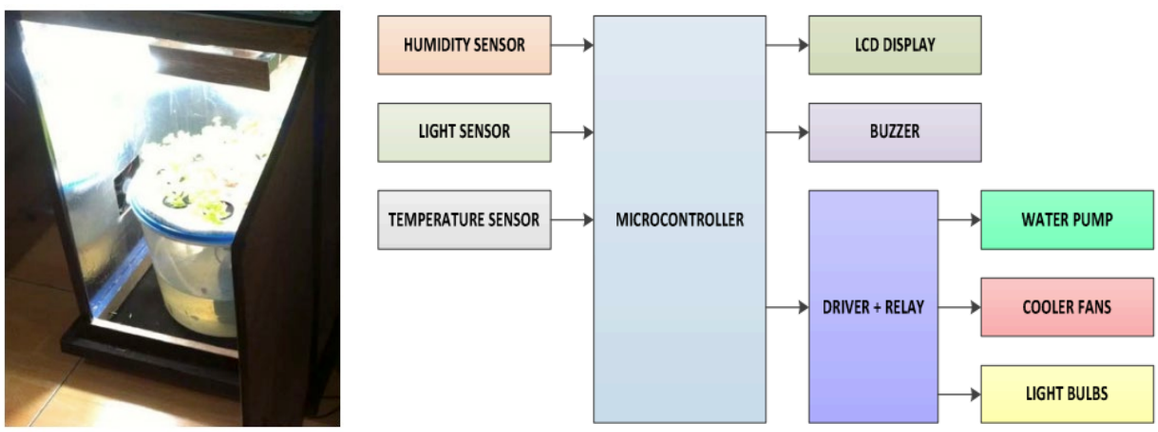
Monitoring and Control of Aeroponic Growing System for Potato Production (Idris and Sani, 2012)



Lastly, Rahmad et al. (2020) developed an automatic aeroponic control system for plant agriculture, as depicted in Figure 7. The system uses a Lattepanda microcontroller interfaced with multiple sensors, such as humidity, light, and temperature, and outputs such as LCD display, buzzer, cooling system, and light bulbs. It can measure and monitor in-situ chamber temperature, like temperature and humidity of cultivated plants, and then can control the nozzle pump, mist maker, fan, and reservoir pump to supply water and nutrients for potatoes. Unfortunately, the irrigation system is found to be inefficient, as installing one requires additional costs and testing.

Figure 7

Automatic Monitoring and Control System in Aeroponic Plant Agriculture (Rahmad et al., 2020)



Furthermore, the researchers considered five relevant patents of similar functionalities or purpose to serve as a reference in defining the technical capabilities of the current study. The following Table 2 lists the relevant patents and their specific features that were adopted and enhanced by the researchers.

Table 2

Related Patents

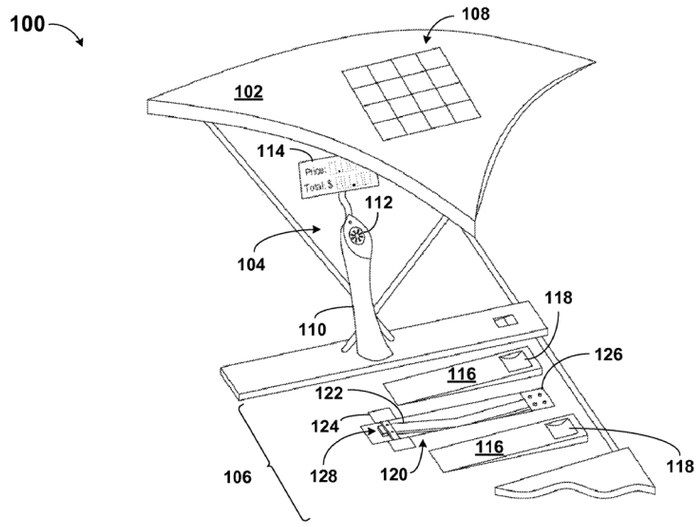
Patent Title	***Author***	***Adopted Feature***	***Enhanced Feature***
Electric vehicle automatic charging station	Dow et al. (2019)	Charging station for e-bikes	Use of RFID card to control the use of the system
Rechargeable battery vending machine	Malaspina (1996)	A vending machine for charging batteries	Interfaced with a microcontroller unit, and other system peripherals to make it for single e-bike
Charging stations for bikes and e-scooters	Anton (2023)	Charging station for e-bikes	Utilized mobile app for monitoring and control of the e-bike
Intelligent Payment Device Based on Radio Frequency Module and Vending Machine	Qie (2021)	RFID-based vending machine	Inclusion of e-payment
Plug-In Electric Vehicle Charging Station Having a Vending Machine Payment Option	Aneesh et al. (2013)	Charging-vending machine station	Inclusion of immobilizer feature

of Purchasing the Item of the Vending Machine while Charging the Rechargeable Electric Vehicle Using Rechargeable Electric Vehicle Interface

The first patent considered is the "Electric Vehicle Automatic Charging Station" by Dow et. al (2019), as shown in Figure 8. It comprises several key components for efficient electric vehicle charging. It includes an alignment mechanism to position the vehicle correctly within the charging station, aided by visual indicators or optical sensors. The coupler assembly, equipped with a charging coupler, is connected to the vehicle's receptacle automatically via an electromagnetic sensor, enabling both physical and electrical connection for battery charging. The charging coupler is replaceable, ensuring longevity and maintenance ease. A cleaning device maintains the charging coupler and receptacle cleanliness. Additionally, it has a terminal with a display, interface, and wireless communication device that facilitates communication and detection of the vehicle, providing specific charging parameters. These parameters, received wirelessly, may include vehicle details such as manufacturer, model, battery information, and installation date, optimizing the charging process. The wireless communication device may include an RFID reader for efficient data transfer. The coupler assembly's horizontal movement allows adaptability to various vehicle sizes, ensuring compatibility across different electric vehicles.

Figure 8

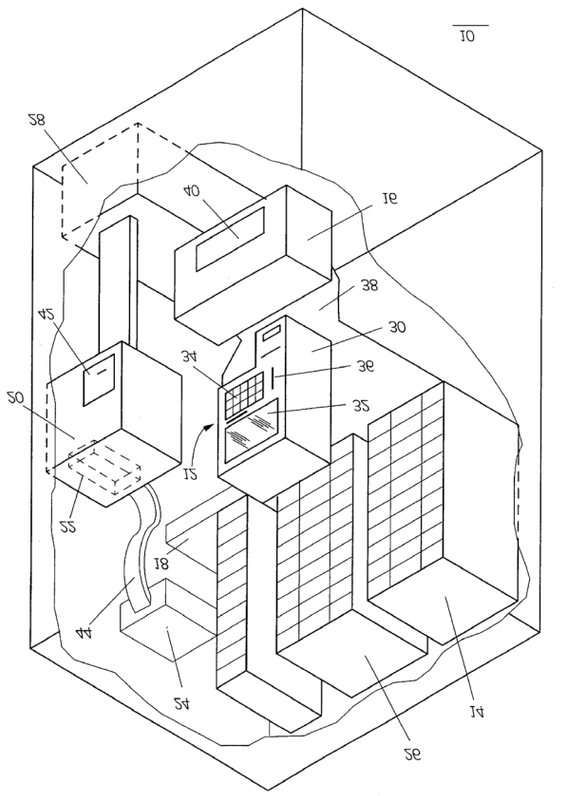
Electric Vehicle Automatic Charging Station (Dow et al., 2019)



Meanwhile, Figure 9 shows the "Rechargeable Battery Vending Machine" by Malaspina (1996), another reference used in this current study. The patent describes a vending machine designed for dispensing rechargeable battery packs and includes various components to facilitate its operation. It consists of a control unit with instruction code stored in its memory, a battery pack receiver unit for accepting and testing received battery packs, a recharging unit for recharging the batteries, and a battery pack handling mechanism for moving the batteries within the machine. Additionally, it includes a money handling unit for processing payments, an interface unit for user interaction, and a dispensing unit for delivering the battery packs. The machine can test the battery packs' voltage levels, assign grades based on test results, and even recondition batteries using a discharge regulator. It also offers various payment options, including coin slots, bill slots, and card readers. The interface unit typically comprises an electronic display, possibly a touch screen for user convenience. Moreover, the vending machine may be connected to a telephone line for two-way communication, enhancing its functionality and management capabilities.

Figure 9

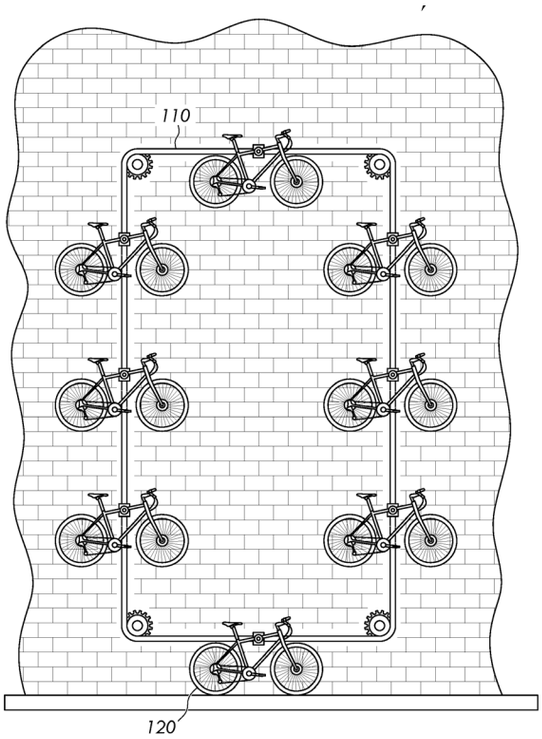
Rechargeable Battery Vending Machine (Malaspina, 1996)



The third patent considered is the "Charging Stations for Bikes and E-scooters" by Anton (2023), as illustrated in Figure 10. It is a parking and charging station for electric bicycles and scooters, featuring a vertical conveyor connected to a power source and communication network. It has horizontal connecting bars, which are attached to both the conveyor and a carrier bar, including universal mounts that offer locking, charging, and data transfer functions. These mounts can accommodate electric bicycles or scooters in various mounting configurations, such as vertical installations, underground setups, or partially above-ground structures. Power sources may include hard wiring to the grid, solar panels, or battery storage. Smart capabilities enable communication with devices, and motorized sprockets rotate the conveyor. Universal mounts can recognize bar codes or RFID chips and detect weight. Another design features a vertical post on a platform with horizontal bars and universal mounts for locking and charging. Additional features include grid wiring, solar panels, battery storage, smart capabilities, and closed-loop bars.

Figure 10

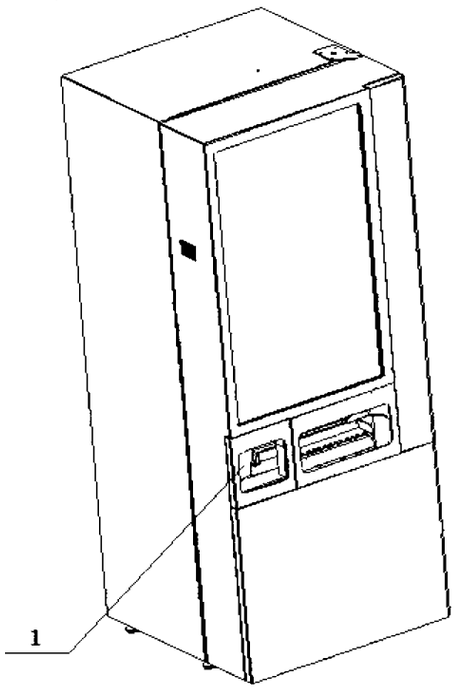
Charging Stations for Bikes and E-scooters (Anton, 2023)



The researchers also considered the "Intelligent Payment Device Based on Radio Frequency Module and Vending Machine" by Qie (2021), as illustrated in Figure 11. It outlines an intelligent payment device and an automatic vending machine based on radio frequency technology. The device includes a system control board, an RFID radio frequency module, and a radio frequency chip. The system control board is linked to the payment management system via a network, and the radio frequency chip is attached to a container carried by the individual collecting goods. Additionally, the system control board integrates Ethernet and serial interface modules, while the RFID module utilizes specific modules like LAN 8720 A and PN 7120. The container can be a shopping basket or cup, with the radio frequency chip embedded within it. In the case of an automatic vending machine, the cup is placed on a supporting table, aligning its radio frequency chip with the RFID module for easy identification.

Figure 11

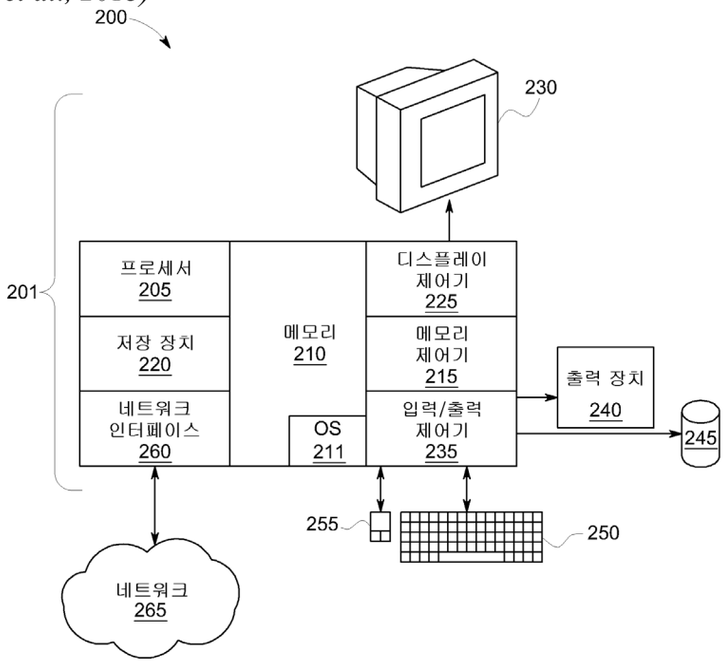
Intelligent Payment Device Based on Radio Frequency Module and Vending Machine (Qie, 2021)



Lastly, the researchers considered the "Plug-In Electric Vehicle Charging Station Having a Vending Machine Payment Option of Purchasing the Item of the Vending Machine while Charging the Rechargeable Electric Vehicle (REV) Using Rechargeable Electric Vehicle Interface" by Aneesh et al. (2013), as shown in Figure 12. The system incorporates an REV interface responsible for processing payments for both electric charging services and vending machine purchases. Additionally, it is equipped with an electrical grid interface for managing energy distribution. This system generates authentication codes upon receiving payment for vending items, which can be encoded onto various media such as charge cards or receipts. The authentication process ensures the validity of transactions. Furthermore, the system extends to encompass the REV itself, which includes a charging station and communication with the vending machine for dispensing prepaid items. A computer program product is also detailed, outlining instructions for facilitating REV charging and vending transactions, including the generation and encoding of authentication codes. These codes are linked to payment methods such as charge cards, streamlining the purchasing process. The program allows for flexibility in payment options and authentication methods, promoting efficient and secure transactions within the REV charging ecosystem.

Figure 12

Plug-In Electric Vehicle Charging Station Having a Vending Machine Payment Option (Aneesh et al., 2013)



Overall, the researchers adopted the features of an electric charging system for e-bikes, vending-type payment, and the use of RFID to control the system. More so, they enhanced these features by integrating them into a compartmentalized system, good for charging one e-bike at a time. Also, the proposed system of the current study has a mobile application for monitoring and control. It also features an alternative e-payment on top of the conventional coin-slot machine and an immobilizer for the security of the e-bike while charging.

III. METHODOLOGY

This section includes the designs, frameworks, materials, methods, procedures, and processes used in realizing the objectives of the capstone study. It also describes what resources, activities, strategies, or techniques that were conducted to translate the design into a tangible output. This section should be written in **detail** to establish the reproducibility and the internal and external validity of the capstone study for the reference of other researchers and the scientific community.

3.1 Prototype/Project/Product/System Design - This section elaborates on the major components of the prototype/project/product/system, including its key features, characteristics, and functionalities. It aims to answer the "design", "formulate," or "characterize" objectives. The illustrations and **detailed** discussions may include the following:

a. **Block Diagram** – a functional connection of the input, process, and output components or materials of the prototype/project/product/system
b. **Pictorial Diagram** – includes the different views of the prototype/project/product/system, such as top, front, isometric, or perspective

with dimensions, including 3-dimensional or Computer-Aided Design (CAD) models

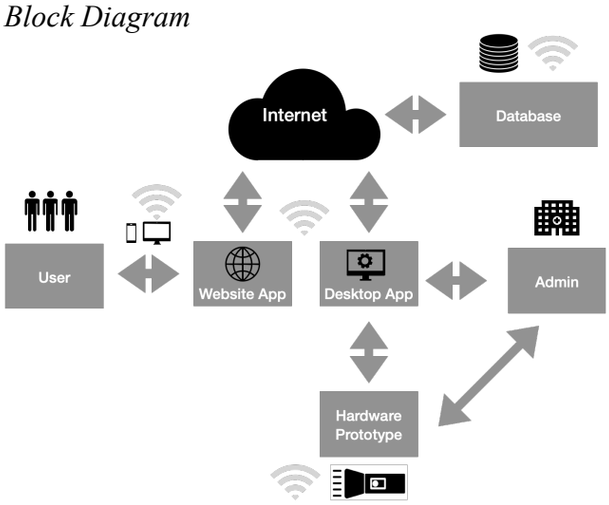
c. **Product / Formulation Mix –** discussion on the mix or optimal ratio design/s as applied in product or materials science, including mathematical formulas or equations
d. **Schematic Diagram –** an electronic or electrical circuit diagram of the hardware components; too detailed schematic diagram can be transferred to the appendices
e. **Algorithm, Framework, and/or Model** – specifications of the algorithm and model/s used in software development, including mathematical formulas or equations
f. **Operational System Flowchart** – this describes how the prototype/project/product/system will function; it can be contextualized depending on the applicability, such as a context diagram, use-case diagram, GUIs, database diagram, etc.

Example:

The following Figure 13 illustrates the overall design of the proposed system. In terms of its software dimension, a patient or the public can access the website application through his/her smartphone or computer to view the availability of hospital beds in nearby locations. Meanwhile, hospital staff can access the admin module through a dedicated desktop application, pre-installed in a specific hospital. The desktop app is based on the Create, Read, Update, and Delete (CRUD) processes for database management, which can manage the availability of hospital beds in direct communication with the hardware prototype through the IoT infrastructure.

Figure 13

Block Diagram



The system maintains a dedicated database that serves as a repository of all processed data for both web and desktop applications. Depending on the information requested by the user, the database collects the data from the desktop application and displays it to the web application. As illustrated in Figure 13, the database is composed of 2 tables, namely “admins” and

“hospitals”. The “admins” table includes the login credentials, such as username and password, as well as the user ID and hospital ID. Meanwhile, the table “hospitals” contains the demographics of various hospitals enrolled in the system, such as hospital ID, hospital name, type, bed admission type, as well as latitude and longitude of the hospital address, contact number, and system settings. Moreover, the system has a hardware prototype, which is divided into two parts – the lower part, which includes the battery holder and nameplate slot, and the upper part, which includes the microcontroller unit (MCU) and 2 optical sensors.

Figure 13

Database Diagram



Figure 14

Hardware Prototype Pictorial Model

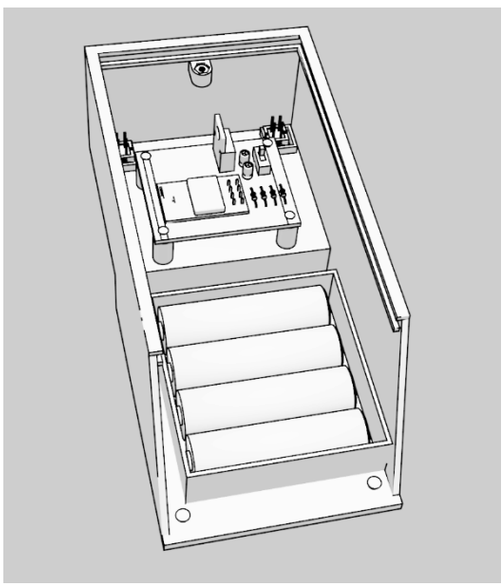
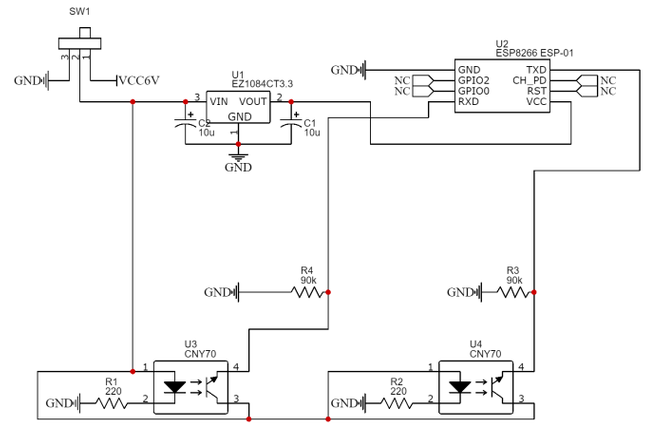


Figure 14 shows the three-dimensional, pictorial model of the hardware prototype. The enclosure was designed to be slightly elevated to optimize the detection consistency. On the other hand, Figure 15 shows the schematic diagram of the hardware prototype.

Figure 15

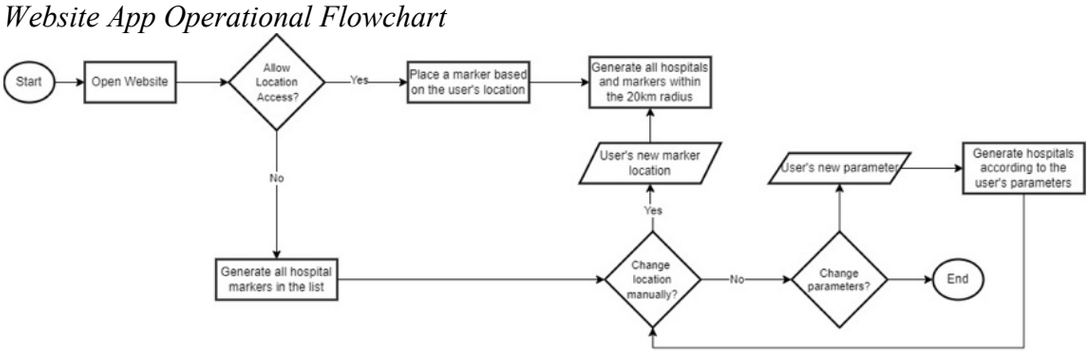
Schematic Diagram



Meanwhile, the following Figure 16 details the operational flowchart of the web app of the system. It provides guidance for the public, hospital staff, and technical support on how to manipulate the web app towards the effective operationalization of the system.

Figure 16

Website App Operational Flowchart



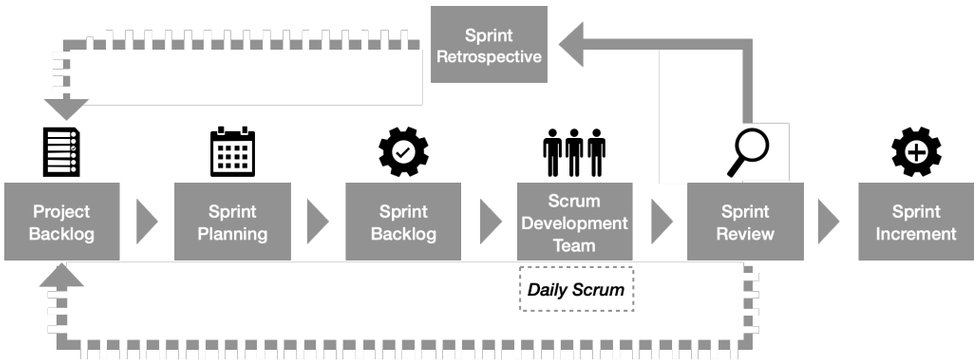
3.2 **Prototype/project/product/system Development -** This section elaborates the procedures, processes, frameworks, or strategies conducted to realize the “develop/ formulate/ fabricate objectives”. Here, the students are expected to project a **single** framework or methodology, depending on the nature of the capstone study. Examples of these are not limited to prototyping, waterfall, agile, scrum, V model, etc. for prototype/project/system development; stage gate food development, experimental, agile, HACCP, mixed methods, etc., for food/product development; and/or experimental, prototyping, engineering process, etc., for material science development.

Example:

The researchers utilized the Agile Scrum methodology as the basis of the development process, as shown in Figure 17. The framework covers both the software and hardware dimensions of the system.

Figure 17

Agile Scrum Methodology



Initially, the researchers contributed to the project backlog by specifying the objectives and the knowledge, software, and hardware requirements of the project. Then, they conducted sprint planning to detail the objectives, related studies, patents, methodologies, and testing and evaluation procedures. Also in this phase, the authors agreed on the specific hardware and software technologies, frameworks, and feature set. Next, they divided the tasks among all members of the team depending on their strengths and capabilities. They also agreed on the timelines, milestones, financial costs, procurement details, communication scheme, risks, and contingency plans. Afterward, the authors prepared the sprint backlog, which lists the specific tasks, deliverables, in charge, and target deadlines for each objective.

Afterwards, they carefully executed the objectives in coordination with technical experts, hospital authorities, and beneficiaries. Initially, they prepared the designs of the hardware, followed by the software dimensions of the system, and conducted iterative development processes for both platforms. For every phase of the development process, the authors reviewed each output to ensure conformity with the project specifications. The hardware prototype was designed to have 2 optical sensors, in which each sensor was configured to represent one bed or a

total of 2 beds in a room. The authors programmed each optical sensor to detect the availability of each bed through a white nameplate. The details of the sprint review or testing are discussed further in the next section – Testing/Analysis Procedure.

Lastly, in terms of the sprint retrospective phase, the researchers involved 50 patients, the public, and another 10-hospital IT staff and administrators to simulate and evaluate the system. They served as pilot respondents for whom their comments and suggestions were captured toward the improvement of the system.

Example:

> **3.3 Testing/ Analysis Procedure -** This includes a detailed discussion of how the "test" or "analyze" objective is realized. This is to ensure that the technical characteristics, e.g., functionality, accuracy, timeliness, reliability, efficiency, nutrient content, tensile strength, etc., are verified and validated. Testing should be quantitative and based on statistical analysis, probability theory, and other scientific numerical analysis.
>
> Formulas and equations used should be presented in the following format (e.g., Cambria Math font), where the number on the right indicates its chronological order as discussed in the text.

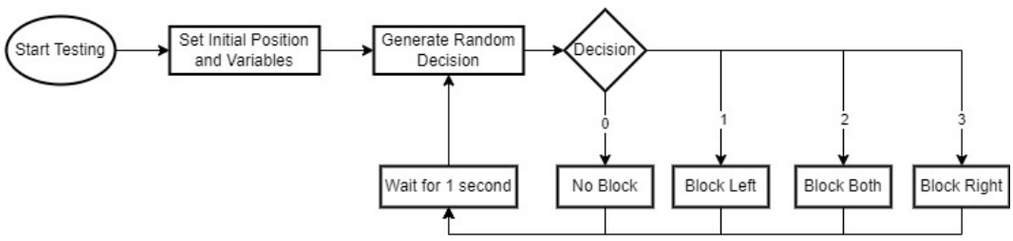
$$a = \frac{(tp + tn)}{(tp + fp + fn + tn)} \quad (1)$$

Example:

As for the testing, the following Figure 18 depicts the testing contraption flowchart for the system. The researchers configured each optical sensor such that once blocked, it would mean that a specific bed in a room is not available. Specifically, they programmed each sensor with "no block", "block left", "block right", and "block both" configurations to indicate "no bed available", "1 bed on the left is available", "1 bed on the right is available", and "both 2 beds are available", respectively. With test rooms 1, 2, 3, 4, and 5, the authors varied the distance readings to 1mm 5mm, 10mm, 15mm, and 16mm, respectively. The responses of the sensors were captured automatically using a script and were collated in CSV format for further statistical analysis.

Figure 18

Testing Contraption Flowchart



Subsequently, the researchers tested the null hypothesis (H_0) of the study, which states that "there is no significant difference in the functional accuracy of the system in sample hospital rooms" with a 95% confidence level and 5% error. They utilized the one-way Analysis of Variance (ANOVA) technique based on the sample size of 5 hospital nameplates with 98 test cases in each room. The analysis was conducted using a one-tailed test through the statistical application software Statistical Package for the Social Sciences (SPSS) version 25. The F-statistic was computed using the following equations, where: $Fs = F\ statistic$; $MSTbg$ = Mean square per treatment between groups; $MSEwg$ = Mean square per error within groups; Λ_{xy} = an observation; $\prod_x$ = group total; Gt= grand total; o_x = number in group x; and o = total observations:

$$Fs = \frac{MSTbg}{MSEwg} \tag{1}$$

$$MSTbg = \sum_{x=1}^{z} \left(\frac{\prod_x^2}{Ox}\right) - \frac{Gt^2}{O} \tag{2}$$

$$MSEwg = \sum_{x=1}^{z} \sum_{y=1}^{z} \Lambda_{xy}^2 - \sum_{x=1}^{z} \left(\frac{\prod_x^2}{Ox}\right) \tag{3}$$

IV. RESULTS AND DISCUSSION

4.1 Prototype/Project/Product/System Description and Structure - This section discusses the finished output and its overall function. It shows the actual image of the prototype/product/product as a whole with appropriate labels**.** This also includes a detailed illustration and discussion of the sections, components, formulation, properties, capabilities, and other key features. It should answer the "design/formulate/characterize" and/or "develop/formulate/fabricate" objectives.

Example:

In this capstone study, the researchers developed a web, desktop application and a hardware prototype that aim to monitor the availability of inpatient beds in hospitals. The following Figures 19 and 20 depict the actual external and internal pictures of the hardware prototype, respectively. The casing is made of a 3D-printed material, with dimensions of 55.5 mm in height, 108.5 mm in length, and 68 mm in width. Beneath the casing is a slightly elevated panel with 17x53x61mm detection space, sufficient to sense the presence of a nameplate. A nameplate is manually inserted by hospital staff into a designated slot to indicate the presence of a patient in a particular bed within a room. The system can detect up to 2 beds or 2 patients per hospital room.

Figure 19

External View

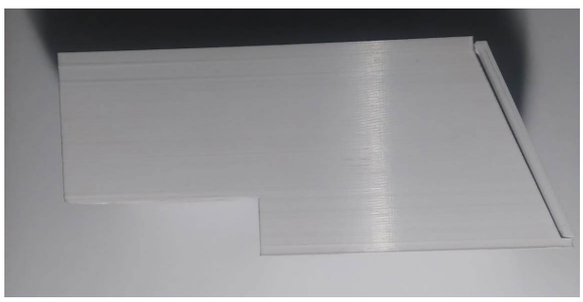


Figure 20

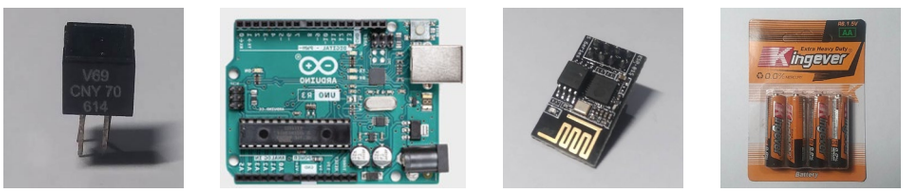
Internal View



Shown in Figure 21 are the main components of the hardware prototype, namely: (a) 2-CNY70 reflective optical sensors, which can detect the presence of the hospital room nameplate indicating that a particular bed is occupied or not; (b) Arduino UNO MCU R3, which includes the ATmega328P processor that controls the operations of the system; (c) ESP8266 ESP-01 Wi-Fi module, which is used to connect to the internet and sends bed availability data to the online database and updates the desktop and website applications used by the general public and admin, respectively; (d) 4 piece-1.5V batteries used to power the whole hardware prototype. These hardware and software components are interconnected through an IoT architecture.

Figure 21

Hardware Components



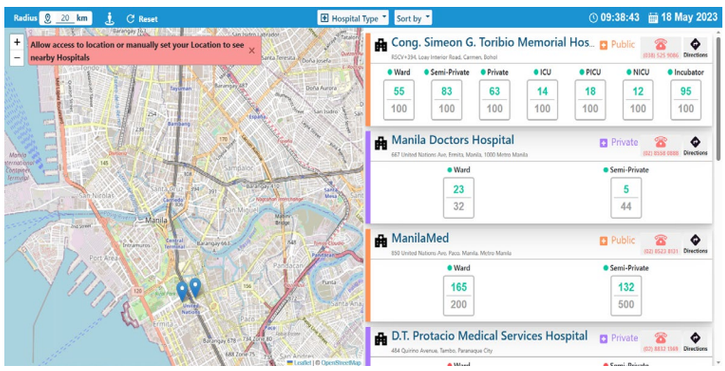
(a) *(b)* *(c)* *(d)*

In terms of the software dimensions, the system operates into 2 platforms – web and desktop applications. Illustrated in Figure 22 is the screenshot of the website application. The authors designed the web application using Figma for the user interface and developed it using Hyper Text Markup Language (HTML), Cascading Style Sheets (CSS), JavaScript, Structured Query Language (SQL), Hypertext Pre-processor (PHP), and Asynchronous JavaScript XML (AJAX). The app features an interactive map through the Leaflet Application Programming Interface (API) and other programming techniques and algorithms, which allow the user to view, zoom, drag, or pan the location of the nearest hospital with the greatest number of available beds. Moreover, the app shows hospitals based on the current location and allows users to manually view specific locations based on latitude and longitude. The app also provides alternatives to users depending on personal preferences, such as public or private hospitals, room types such as ward, semi-private, or private, as well as contact details of a specific hospital. The app also

works offline using AJAX and automatic location tracking using Geolocation API, and it features a Nearest Location Finder algorithm, programmed using SQL and PHP, which expedites the querying and location tracking processes.

Figure 22

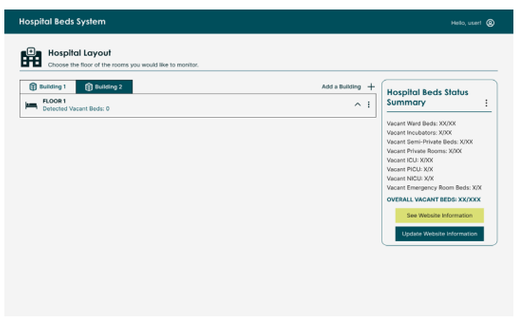
Website Application (Manila Area)

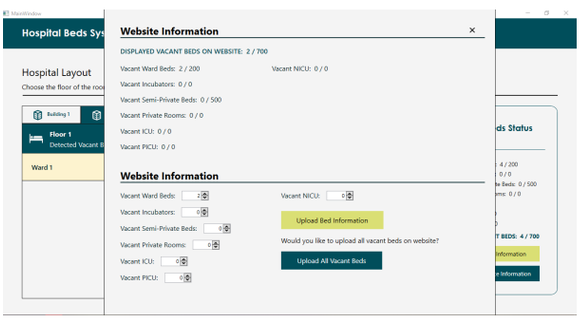


Meanwhile, the desktop application of the system, as shown on Figure 23 supports both the hardware and the central database. The authors utilized Figma for the user interface and developed the application using the Microsoft #NET framework and Windows Presentation Foundation. The desktop is intended for hospital staff, which allows them to monitor and update the availability of hospital beds based on the data fed by the hardware prototype. It features real-time querying updates, configured using C# programming language. The results are then subsequently displayed on the web application for the information of the public. Shown in Figures 17a, 17b, and18 are the screenshots of the desktop application's homepage, modification page, and the C# Programming language implementation, respectively.

Figure 23

Desktop App





4.2 Test/ Analysis Results - This section discusses the detailed performance results and capabilities, including the limitations, of the prototype/project/product/system, based on the "test/analyze" objectives and procedures conducted. The salient findings can be presented in themes, following the pre-determined test/analysis parameters. For product/materials development, results may include laboratory results from regulatory bodies, testing centers, etc.

The results can be presented using tables and figures, such as scatter plots, histograms, pie charts, trends, matrices, heat maps, etc. The results of testing/ analysis should be compared, related, correlated, verified, or validated with other empirical studies, similarly published prototypes/projects/products/systems, patented technologies, industry and/or international standards, etc. The results can also be modeled through forecasting, categorization, prediction techniques, and the likes.

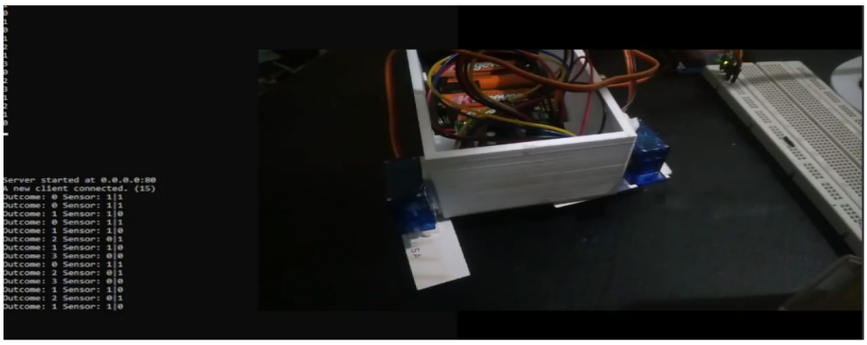
The prescribed format for figures and tables is similar to those used in the "Literature Review" section, except for the cited author/s. Lastly, test or analysis results should be interpreted using various statistical techniques, hypothesis testing approaches, probability theory, and other methods of scientific and numerical analyses.

Example:

The researchers then tested the system by focusing on the functional accuracy of the hardware prototype, through its reflective optical sensors installed in 5 nameplates. Shown in Figure 24 is the actual testing procedure conducted. Each nameplate was tested for approximately 15 minutes with an interval of less than a second.

Figure 24

Testing Setup with Screenshot



The study found that the true detection range of the hardware prototype is from 1mm to a maximum of 15mm only. Unfortunately, 16mm yields to no data detection, hence, the authors excluded it in further testing the H_0. The following Table 1 shows the summary of testing from the sample test cases. In this scenario, only 12 out of 98 test cases per set of 1mm, 5mm, 10mm, 15mm, and 16mm detection readings.

Table 3

12 over 98 Sample Test Case Results

No.	*1mm*		*5mm*		*10mm*		*15mm*		*16mm*	
	Expected	*Actual*	*Expected*	*Actual*	*Expected*	*Actual*	*Expected*	*Actual*	*Expected*	*Actual*
87	3	3	3	3	0	0	2	2	1	NaN
88	0	0	0	0	3	3	3	3	2	NaN
89	2	2	2	2	1	1	1	1	3	NaN
90	3	3	3	3	0	0	0	0	2	NaN
91	1	1	1	1	2	2	3	3	1	NaN
92	3	3	3	3	3	3	2	3	0	NaN
93	2	2	2	2	2	2	0	0	2	NaN
94	3	3	3	3	1	1	2	2	0	NaN
95	2	2	2	2	3	3	1	1	3	NaN
96	0	0	0	0	0	0	3	3	2	NaN
97	2	2	2	2	1	1	2	2	1	NaN
98	3	3	3	3	2	2	1	1	2	NaN

Furthermore, shown in Table 4 is the result of the inferential statistics. The researchers found that there is no statistically significant difference in the detection accuracy of the 4 optical sensors determined by the one-way ANOVA ($F(3, 388) = 1.00$, $p=0.393$). Hence, they accepted the H_0. This means that the optical sensors are consistent and at par with each other in detecting accurately if a specific hospital room is available, partially available, or not available.

Table 4

ANOVA Test Results

	Sum of Squares	*df*	*Mean Square*	*F*	*Sig.*
Between Groups	0.008	3	0.003	1.000	0.393
Within Groups	0.990	388	0.003		
Total	0.998	391			

V. CONCLUSIONS

This section includes discussions of what the capstone study is about in a nutshell. It also includes the significant findings that answer the problem, research gap, and objectives of the capstone. Pertinent insights, realizations, and learnings should also be included to provide inputs to the scientific body of knowledge and the community of researchers, scientists, and other stakeholders.

Similarly, the limitations or constraints of the prototype/project/product/system, including the recommended actions for future research should be presented. It should also capture the comments and suggestions from the end-users, advisory committee, and other interested parties.

Example:

The proposed system aims to contribute to a more efficient bed management and decision-making process among patients and hospital managers alike. Specifically, the website application can provide real-time and accurate information on bed availability in hospitals through automatic location tracking or a manual query. It provides an interactive map that allows users or the public to view the navigational route to the selected hospital, including contact details, making it more convenient for communication purposes. Meanwhile, the desktop app allows the hospital admin staff to monitor and update bed vacancies based on the data fed by the hardware prototype through the IoT architecture. It has a secure login system and it has a provision for adding or removing device keys to recognize or unrecognize nameplates. Moreover, the use of the IoT infrastructure, locally available materials, and the agile scrum methodology are deemed effective in attaining the study's objectives. Test results show that the system is consistently accurate in detecting the availability of hospital beds. Hence, the system is considered a viable solution to ease the overcrowding of hospitals and a proactive communication tool for potential inpatients or the public. The outcomes of the system are expected to contribute to the UN-SDG-3 on good health and well-being. Future research should focus on applying the system to more hospitals at the regional or national levels to ensure consistency and reliability.

VI. ACKNOWLEDGMENT

This contains the student/s statement of gratitude to the people or entities who have contributed to the completion of the capstone study.

Example:

The researchers would like to thank the voluntary participation of hospital administrators and staff from Capitol Medical Center, Medical Center of Muntinlupa, Commonwealth Health Center, University of Perpetual Help System Dalta, Philippine General Hospital, and St. James Hospital Inc. Their comments, suggestions, and feedback toward the improvement of the system are earnestly recognized. Likewise, the researchers would like to acknowledge the wisdom and support of the Electronics Engineering Technology Department, College of Industrial

Technology, Technological University of the Philippines- Manila, led by then-Head, Prof. Gemma D. Belga, and the faculty members – Prof. Marc V. Ardie Ardiente, Prof. Dennis J. Tabucol, Prof. Hilda V. Robino, Prof. Karen K. Foronda, and Prof. Javi Ann O. Interno (+).

VII. DECLARATION OF CONFLICT OF INTEREST

This section aims to establish that the capstone study has been conducted objectively and ethically. As such, the student/s should disclose any personal or professional relationship/s to other people or entities, such as funding agencies, respondents, testing centers, regulatory bodies, university administrators, publishers, sponsors, etc., that may have direct or indirect influence on the results of the study. The extent of use of AI tools should be declared.

Moreover, the student/s should provide information on compliance with ethics review or clearance requirements, particularly on the use of humans, animals, and the environment as subjects of the capstone study. They should ensure that they have abided by all applicable legal requirements, such as data privacy, binding contracts, internal policies, and the like.

Example:

The researchers declare that there is no conflict of interest among themselves as authors, no external funding has been sought, no use of human, animals, or the environment as "subjects", and no Artificial Intelligence (AI) tools have been used during the design, development, and testing of the capstone study.

VIII. REFERENCES

The reference list provides a way for readers to locate and verify the works of cited authors. References are used to document and substantiate statements made about the literature, just as data in the paper are used to support interpretations and conclusions. The references cited in the paper do not need to be exhaustive but should be sufficient to support the need for the capstone study and to enable readers to place it in the context of previous research. Only authors cited in the manuscript should be included in the reference list.

References should follow the most recent American Psychological Association (APA) format, e.g., APA 7th edition, 2020. References should be arranged alphabetically, single-spaced, and the second line onwards should be hang-indented by 0.5 inches. Student/s may use the paragraph formatting function of the word processing application to effect this hanging-indent format.

Use "et al." in the reference list if the total authors exceeds twenty (20), otherwise cite all the names of the 20 authors.

The following are sample APA v7 formats of refereed sources:

1. Journal Article with DOI

Lastname, F. M., & Lastname, F. M. (Year). Title of article. Title of Periodical, Vol. (Issue), page numbers. DOI or URL

Example:

Hendy, P., Patel, J., Kordbacheh, T., Laskar, N., & Harbord, M. (2012). In-depth analysis of delays to patient discharge: a metropolitan teaching hospital experience. *Clinical Medicine*, *12*(4), 320–323. https://doi.org/10.7861/clinmedicine.12-4-320

2. **Journal Article with no DOI**

Lastname, F. M., & Lastname, F. M. (Year). Title of article. Title of Periodical, Vol. (Issue), page numbers.

Example:

Mury, T., & Fusco, V. F. (2007). Inverse class-E amplifier with transmission-line harmonic suppression. *IEEE Transactions on Circuits and Systems I: Regular Papers*, *54*(7), 1555-1561.

3. **Book**

Lastname, F. M. (Year). Title of book. Publisher. DOI or URL

Example:

Fellows, P. J. (2022). Food processing technology: principles and practice 5th edition. Woodhead publishing. https://www.sciencedirect.com/book/9780081019078/food-processing-technology

4. **Book Chapters**

Lastname, F. M. (Year). Title of chapter. In F. M. Lastname of Editor (Ed.), Title of book (page numbers). Publisher. DOI or URL

Example:

Pacheco-Torgal, F., & Jalali, S. (2011). Nanotechnology in eco-efficient construction. In F. Pacheco-Torgal, M. Cabeza, J. Labrincha, & A. F. Castro-Gomes (Eds.), Eco-efficient concrete (pp. 337–362). Woodhead Publishing. https://www.sciencedirect.com/book/9780857097675/eco-efficient-construction-and-building-materials

5. **Conference Proceedings**

Lastname, F.M. (Year, Month day/s). Title of article. Conference name, Location. DOI or URL

Example:

Garg, H., & Dave, M. (2019, April). Securing iot devices and securely connecting the dots using rest api and middleware. In 2019 4th International Conference on Internet of Things: Smart Innovation and Usages (IoT-SIU), Ghaziabad, India. https://doi.org/ 10.1109/IoT-SIU.2019.8777334

6. **Government Websites**

Government Agency. (Year). Title of article. URL

Lastname, F.M. (Year). Title of article. Government Agency. URL

Example:

Department of Science and Technology. (2022). Harmonized national R&D agenda 2022-2028. https://www.dost.gov.ph/knowledge-resources/downloads/file/5669-harmonized-national-r-d-agenda-2022-2028.html

7. **Patent Databases**

Lastname, F.M. (Year Patent Issued). Title of patent (Patent No.). Country Patent Office Name. URL.

Example:

Dow, E., Gilchrist, J., Lewis, B., McMillan, K., & Nycz, C. (2019). Electric vehicle automatic charging station (US10279696B2). US Patent Office. https://patents.google.com/patent/US10279696B2/en?q=(Ebike+Charging+Station)&oq=E-bike+Charging+Station

8. **Reputable Non-Government Websites**

Non-Government Agency. (Year). Title of article. URL

Lastname, F.M. (Year). Title of article. Non-Government Agency. URL

Example:

United Nations Organization. (2025). Sustainable development goals. https://sdgs.un.org/goals

IX. APPENDICES

This section includes pertinent supplementary documents or references used in the capstone study, such as:

1. **Timeline –** Gantt chart, matrix that shows the activities and milestones, etc.
2. **Development Pictures -** actual pictures taken during the development of the prototype/ project/ product/ system; each picture should have captions and dates
3. **Laboratory Test Results –** scanned or digital copies of test results from the laboratory, testing centers, and the like
4. **Research Ethics Review –** a clearance certificate obtained from the URDS or relevant authority on the ethical use of humans, animals, or the environment as subjects, if applicable.
5. **Certificate of Similarity Index -** this document certifies that the capstone study is novel, original, or an improvement or innovation of an existing prototype/project/product/system based on the results of similarity-checking application software. The certificate can be obtained from the URDS or equivalent authority.

6. **Certificate of Professional Editing and Proofreading –** this document certifies that the capstone study is proofread by a qualified technical writer or professional expert, who should be **accredited** by the university, subject to payment of fees in accordance with university/BOR-approved rates.

X. STUDENT'/S' BIONOTE

This section includes a concise description of the credentials of the students/s including their contributions and role in the capstone study, as applicable.

Example:

Jofer S. Usa is a Software Engineer at Samsung Electronics Philippines Corporation. His research interests include software development and optimization, embedded systems, and operating systems. In this study, he developed the desktop application and hardware prototype.

XI. GENERAL GUIDELINES

1. The manuscript paper should be 8.5" x 11" in size, double-spaced, with margins of 1.5 inches on the left and 1.0 on all other sides.
2. The default font style should be Times New Roman, size 12, except for the title of the capstone, which should be size 14.
3. The manuscript should be written in third-person format and should be presented in a scholarly manner.
4. Page numbers should be written in the upper right section of the header, except for the title page. The preliminaries section should be numbered in small Roman numerals.
5. The student/s should use the present tense when discussing figures, tables, facts, or data; the past tense if discussing how a certain process, procedure, activity, or method was done; the future tense if discussing how a certain process, procedure, activity, or method will be done, including the recommendations.
6. The following are the recommended in-line citation formats:
   Parenthetical citation: (Santos et al., 2019)
   Narrative citation: Santos et al. (2019)
7. The student/s may use application software and the like to improve writing accuracy, style, tone, and other best practices, to enhance the quality, reliability, and professionalism of the manuscript.
8. The use of Artificial Intelligence (AI)- driven tools, like ChatGPT etc., may be allowed, provided that their extent of use or contributions to capstone study are declared in Section VII - Declaration of Conflict of Interest. The text generated from these tools should be rewritten or revised based on their contextual use and applicability to the capstone study. Meanwhile, the information/data, figures and tables generated by the AI tools should be cross-validated against the refereed/ published literature, and the original authors should be cited in Section II- Literature Review. The allowable similarity index and/or AI-generated text or data should not be more than 15%.
9. The final hardbound cover page should bear the following information:

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10. For the undergraduate and graduate programs, the hardbound cover color should be "green" and "maroon", respectively. The text of the cover page should be embossed using the standard color "gold".

11. Meanwhile, the side of the hardbound should bear the following information:

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12. The following is a sample format for the "Certificate of Professional Editing and Proofreading":

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13. The following template is shared as a sample reference in preparing the main sections of a capstone study manuscript.